POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2019
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Test whether the given discrete-signal is periodic. If so, find the fundamental period. 7

   i. $x[n] = \sin\left(\frac{2\pi}{3}\right)n$ ii. $x[n] = 4e^{j\frac{4\pi\left(n+\frac{1}{3}\right)}{5}}$

   b) A digital discrete time signal x[n] = 6.5cos(0.1πn) is quantized with the resolution, Δ = 0.01. How many bits are required in the A/D converter? If the maximum frequency that can be reconstructed from above signal is 500Hz, determine the reconstructed time signal? 8

2. a) State and prove the necessary and sufficient conditions for an LTI system to be causal and stable. 7

   b) Consider an LTI system with impulse response h[n] = u[n] – u[n-5] and input x[n] = $(3/5)^n${u[n]-u[n-4]}. Determine the output of the system, y[n]. 8

3. a) Determine the causal signal x[n] if its Z-transform X(z) is given by 7

$$X(z) = \frac{1 + 3z^{-1}}{1 + 3z^{-1} + 2z^{-2}}$$

   b) With the help of N = 8, explain radix-2 decimation-in-time (DIT) FFT algorithm for computation of DFT. Give the computational efficiency of FFT over DFT. 8

4. a) Using circular convolution method, determine the linear convolution of the following sequences: x[n]= $\{\underset{\uparrow}{1}, 2, 4\}$ and h[n] = h[n]= $\{\underset{\uparrow}{1}, 2, 1, 3\}$. 8

   b) What are the advantages of representing the digital filter in the block 7

This document is available on studocu

Available on notes.arpanadhikari7.com.np

diagram form? Draw the direct form structures of the system described by LCCD equation

$$y[n] = 0.3y[n-1] - 0.9y[n-3] + x[n] + 2x[n-1] + 4x[n-2]$$

5. a) Design a digital Butterworth low pass filter whose transfer function is given by

$$\begin{cases} -3.098dB \leq |H(e^{jw})| \leq 0 & 0 \leq w \leq 0.2\pi \\ |H(e^{jw})| \leq -10.46dB & 0.6\pi \leq w \leq \pi \end{cases}$$

Use impulse-invariant transformation.

b) Draw the lattice structure for the following IIR system.

$$H(z) = \frac{1}{3+6z^{-1}+9z^{-2}}$$

Is the system stable?

6. a) Define symmetric and Anti-symmetric filter, and discuss the applications.

b) Design a lowpass filter which will have -3 dB cut-off at 30π rad/sec and an attenuation of 55 dB at 48π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 200 samples/sec.

notes.arpanadhikari7.com.np

7. Write short notes on:
   a) Filter Design by Kaiser Window
   b) Compare and constrast FIR and IIR
   c) Frequency shift property of DFT

Available on notes.arpanadhikari7.com.np